

HYDROGEN & ITS COMPOUNDS

1. Similarity of H with halogens is
 - (1) Forms diatomic molecules
 - (2) form uninegative ions
 - (3) having one 'e' less to acquire stability
 - (4) All
2. If M.P. of Hydrogen is 13.96K then M.P. of deuterium will be
 - (1) 9.8 (2) 13.96 (3) 18.7 (4) not fixed
3. A mixture of H_2 and CO is known as
 - (1) water gas (2) Syn gas
 - (3) Synthesis gas (4) all
4. Reaction $CO_{(g)} + H_{2O_{(g)}} \xrightarrow[Fe]{673K} CO_{2(g)} + H_{2(g)}$ is known as
 - (1) water gas reaction (2) water gas shift reaction
 - (3) Uyenos reaction (4) eletrolytic method
5. dihydrogen is–
 - (1) water soluble (2) water insoluble
 - (3) sparingly soluble (4) Highly water soluble
6. hydrides like MgH_2 , B_2H_6 are known as
 - (1) Nonstoichiometric compound
 - (2) Interstitial compound
 - (3) Binary compound
 - (4) Clathrates compound
7. CH_4 is an example of–
 - (1) 'e' deficient covalent hydride
 - (2) 'e' precise covalent hydride
 - (3) 'e' rich covalent hydride
 - (4) 'e' precise saline hydride
8. Term 'H-economy' means
 - (1) low costing in preparation of H_2
 - (2) fuel efficiency of H_2
 - (3) transportation and storage of energy in gas/liq H_2 form
 - (4) Both 2, 3
9. After ocean water major source of water is
 - (1) Ground water
 - (2) Atmospheric vapours
 - (3) Riverwater
 - (4) Gl i & l i
10. Strong Intermolecular H-Bonding in water leads to
 - (1) High heat of fusion
 - (2) High heat of vaporisation
 - (3) High freezing point
 - (4) All
11. Which relation is true. regarding dielectric constant (ϵ_o)?
 - (1) $\epsilon_{o_{H_2O}} > \epsilon_{o_{D_2O}}$ (2) $\epsilon_{o_{H_2O}} < \epsilon_{o_{D_2O}}$
 - (3) $\epsilon_{o_{H_2O}} = \epsilon_{o_{D_2O}}$ (4) Data not known
12. Crystalline form of water is–
 - (1) Dry ice (2) Ice
 - (3) water liq at 4°C (4) supercooled water
13. At low temp ice crystallises in
 - (1) hexagonal form (2) Cubic form
 - (3) Both (4) None
14. In clark method chemical used is
 - (1) NaOH (2) Na_2CO_3 (3) $CaCO_3$ (4) $Ca(OH)_2$
15. Calgon method is used to remove which type of hardness of water?
 - (1) temporary (2) permanent
 - (3) Both (4) None
16. Which substance is added as stabiliser in storage of H_2O_2
 - (1) dust (2) $CaCl_2$ (3) Ca-silicates (4) Urea
17. Which method is suitable to remove both temporary as well as permanet hardness?
 - (1) Clark method (2) calgon method
 - (3) permutit method (4) 2 and 3 both
18. Perhydrol is an aquated solution of
 - (1) Bleaching powder (2) ozone
 - (3) $KMnO_4$ (4) H_2O_2
19. Not true about H_2O_2
 - (1) Pale Blue liquid
 - (2) two-OH bonds lies in same plane
 - (3) act as both oxidant and reductant
 - (4) stored in wax lined glass and kept in dark
20. Mixture of hydrazine and H_2O_2 is a–
 - (1) antiseptic (2) insecticide
 - (3) Bl h (4) k f l